



Launch and Manage an AWS EC2 Web Server Instance

+ Add Description

Nhlanhla Malaza | 102 steps | 23 minutes

V Vocareum

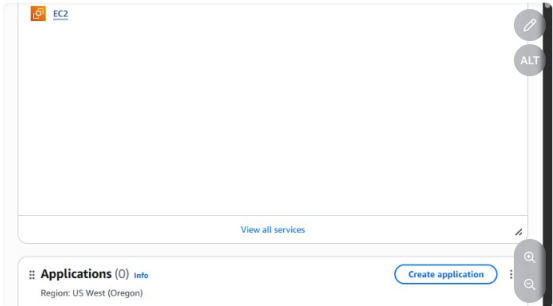
54,188,189,231

Newtab

Amazon Web Services

+

1 Navigate to <https://us-west-2.console.aws.amazon.com/console/home...>

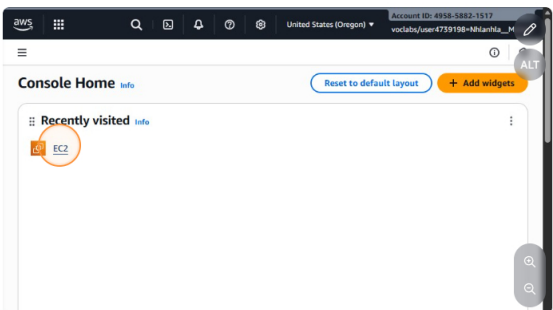


+

2 Switch to tab Workbench - Vocareum"

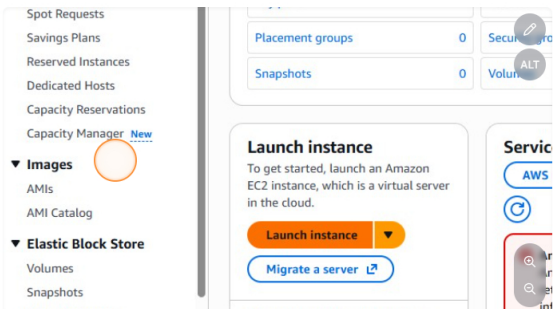
+

3 Click "EC2"



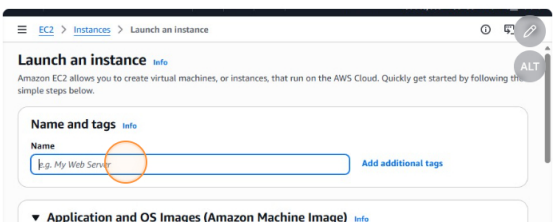
+

4 Click "Launch instance"



+

5 Click the "Name" field.



6 Type "Web Server"

7 Click here.

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Quick Start



Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-055a9df0c8c9f681c (64-bit (x86), uefi-preferred) / ami-0b90f603064208507 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term

8 Click "ami-055a9df0c8c9f681c (64-bit (x86), uefi-preferred) / ami-0b90f603064208507 (64-bit (Arm), uefi)"



Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-055a9df0c8c9f681c (64-bit (x86), uefi-preferred) / ami-0b90f603064208507 (64-bit (Arm), uefi)
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Free tier eligible

Amazon Linux 2023 kernel-6.1 AMI
ami-055a9df0c8c9f681c (64-bit (x86), uefi-preferred) / ami-0b90f603064208507 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Amazon Linux 2023 kernel-6.12 AMI
ami-0fcee47b1475c1af3 (64-bit (x86), uefi-preferred) / ami-0cbac0f1d6260a580 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Deep Learning Base AMI with Single CUDA (Amazon Linux 2023)
ami-03ea4dc10ef1893f8 (64-bit (x86)) / ami-0f9ea93fc561daee2 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Deep Learning OSS Nvidia Driver AMI GPU PyTorch 2.9 (Amazon Linux 2023)
ami-0722d219de962b2ee (64-bit (x86)) / ami-05cc4d4d496ada1f3 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

9 Click here.

Verified provider:

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand RHEL base pricing: 0.0392 USD per Hour
On-Demand Linux base pricing: 0.0104 USD per Hour

Free tier eligible

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name required

10 Click "On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour"

On-Demand Linux base pricing: 0.0116 USD per Hour
On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand RHEL base pricing: 0.026 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour

t2.small
Family: t2 1 vCPU 2 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.023 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.025 USD per Hour
On-Demand SUSE base pricing: 0.053 USD per Hour

t3.micro
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand RHEL base pricing: 0.0392 USD per Hour
On-Demand Linux base pricing: 0.0104 USD per Hour

Free tier eligible

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

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On-Demand Linux base pricing: 0.0104 USD per Hour
Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - **required**

Select

[Create new key pair](#)

▼ Network settings [Info](#)

[Edit](#)

Network [Info](#)

vpc-00b761c040f955598

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

+

12 Click "Proceed without a key pair (Not recommended)"

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - **required**

Select

[Create new key pair](#)

Q |

Proceed without a key pair (Not recommended)

Default value

[Edit](#)

Network [Info](#)

vpc-00b761c040f955598

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

+

13 Click "Edit"

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - **required**

Proceed without a key pair (Not recommended)

Default value

[Create new key pair](#)

▼ Network settings [Info](#)

[Edit](#)

Network [Info](#)

vpc-00b761c040f955598

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

+

14 Click "172.31.0.0/16"

launch the instance.

Key pair name - **required**

Proceed without a key pair (Not recommended)

Default value

[Create new key pair](#)

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-00b761c040f955598

172.31.0.0/16

(default)

Subnet [Info](#)

No preference

▼

[Create new subnet](#)

Availability Zone [Info](#)

No preference

▼

[Enable additional zones](#)

Auto-assign public IP [Info](#)

Enable

▼

Firewall (security groups) [Info](#)

+

15 Click "vpc-08f793284fe44ea0c (Lab VPC)"

Proceed without a key pair (Not recommended)

Default value

[Create new key pair](#)

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-00b761c040f955598

172.31.0.0/16

(default)

Q

vpc-08f793284fe44ea0c (Lab VPC)

10.0.0.0/16

[Create new subnet](#)

vpc-00b761c040f955598

172.31.0.0/16

(default)

✓

[Enable additional zones](#)

Auto-assign public IP [Info](#)

Enable

▼

16 Click the "Security group name -required" field.

Availability Zone: us-west-2b (usw2-az2) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.0.2.0/24

Create new subnet

Auto-assign public IP | Info

Enable

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Security group name - required

launch-wizard-1

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./0#,@[]+=&:()!\$*

Description - required | Info

launch-wizard-1 created 2026-01-26T12:49:17.352Z

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type | Info Protocol | Info Port range | Info

Remove

17 Type " Backspace Web Server security group"

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Security group name - required

Web Server security group

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./0#,@[]+=&:()!\$*

Description - required | Info

launch-wizard-1 created 2026-01-26T12:49:17.352Z

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type | Info Protocol | Info Port range | Info

ssh TCP 22

Source type | Info Source | Info Description - optional | Info

Anywhere Add CIDR, prefix list or security group e.g. SSH for admin desktop

Remove

19 Type "Security group for my web server"

Security group name - required

Web Server security group

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./0#,@[]+=&:()!\$*

Description - required | Info

Security group for my web server

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type | Info Protocol | Info Port range | Info

ssh TCP 22

Source type | Info Source | Info Description - optional | Info

Anywhere Add CIDR, prefix list or security group e.g. SSH for admin desktop

0.0.0.0/0

Remove

21 Click the "8" field.

Inbound Security Group Rules

No security group rules are currently included in this template. Add a new rule to include it in the launch template.

Add security group rule

► Advanced network configuration

▼ Configure storage | Info Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Add new volume

Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems

22 Click here.

[Add new volume](#)

Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems [Edit](#)

► **Advanced details** [Info](#)

▼ **Summary**

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10....[read more](#)
ami-055a9df0c8c9f681c

23 Click "Select"

Select ▼

Shutdown behavior [Info](#)
Stop ▼

Stop - Hibernate behavior [Info](#)
Select ▼

Termination protection [Info](#)
Select ▼

Stop protection [Info](#)
Select ▼

Detailed CloudWatch monitoring [Info](#)
Select ▼

Credit specification [Info](#)
Unlimited ▼

24 Click "Enable"

Shutdown behavior [Info](#)
Stop ▼

Stop - Hibernate behavior [Info](#)
Select ▼

Termination protection [Info](#)
Select ▲
Select ✓
Enable
Disable

Detailed CloudWatch monitoring [Info](#)
Select ▼

Credit specification [Info](#)
Unlimited ▼

CloudShell Feedback Privacy Terms Cookie pre...

25 Click the "User data - optional" field.

aws

United States (Oregon) ▼

EC2 > Instances > Launch an Instance

User data - optional [Info](#)
Upload a file with your user data or enter it in the field.
[Choose file](#)

26 Press **ctrl** + **a**

User data - optional [Info](#)
Upload a file with your user data or enter it in the field.
[Choose file](#)

27 Click the "User data-optional" field.

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
yum -y install httpd
systemctl enable httpd
systemctl start httpd
echo '<html><h1>Hello From Your Web Server</h1></html>' >
/var/www/html/index.html
```

☐ User data has already been base64 encoded

28 Click "Launch instance"

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10.....[read more](#)
ami-055a9d0c8c9f681e

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

29 Click "Successfully initiated launch of instance (i-004279c2214babbb3)"

Success
Successfully initiated launch of instance (i-004279c2214babbb3)

Launch log

Next Steps
What would you like to do next with this instance, for example "r"

Create billing usage alerts
To manage costs and avoid surprise bills, set up email

Connect to your instance
Once your instance is running, log into it from your local computer.

Connect an RDS database
Configure the connection between an EC2 instance and a database to

30 Click "View all instances"

Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
[Create EBS snapshot policy](#)

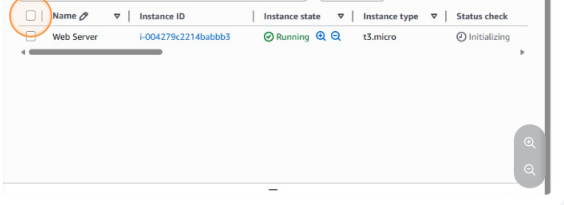
Manage detailed monitoring
Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.
[Manage detailed monitoring](#)

Create Load Balancer
Create an application, network gateway or classic Elastic Load Balancer
[Create Load Balancer](#)

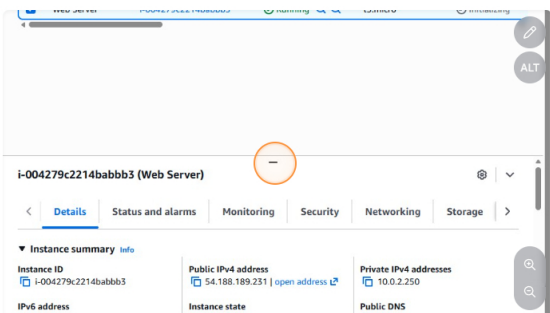
[View all instances](#)

31 Click this checkbox.

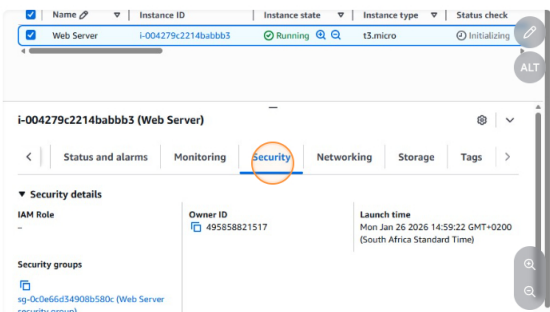
EC2 > Instances



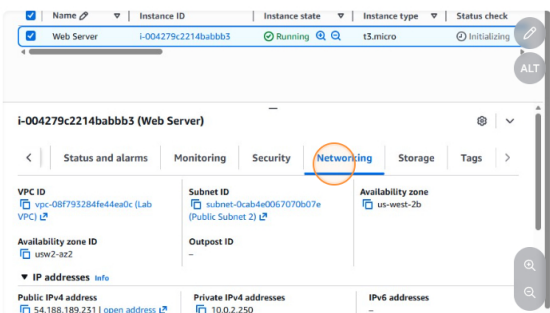
32 Click "Slider"



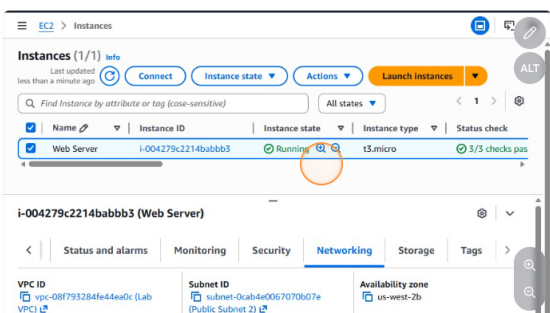
33 Click "Security"

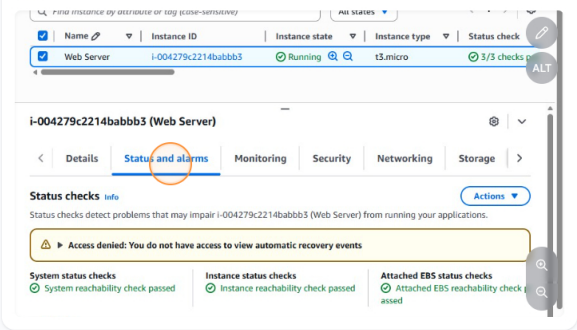


34 Click "Networking"

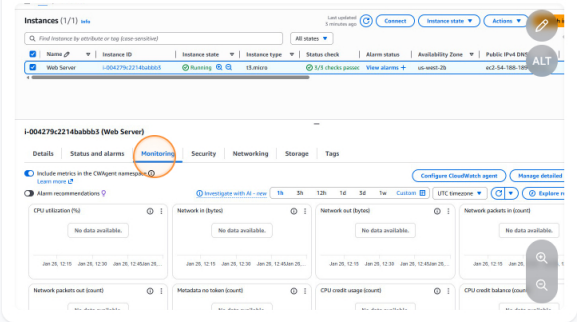


35 Click here.

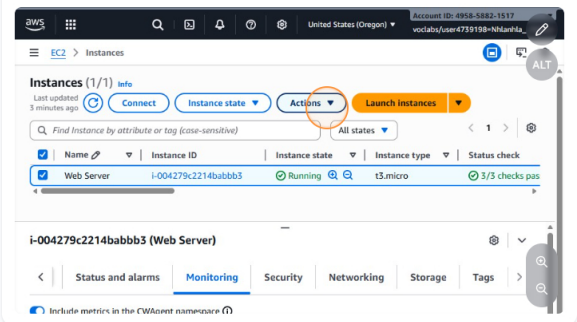




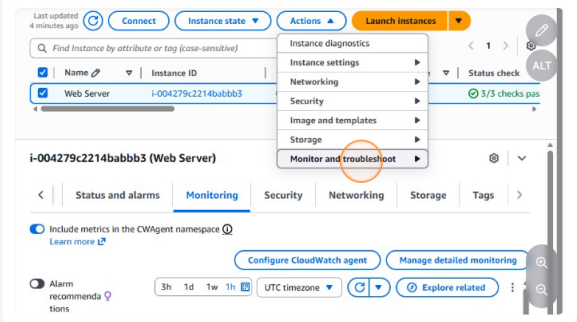
37 Click "Monitoring"



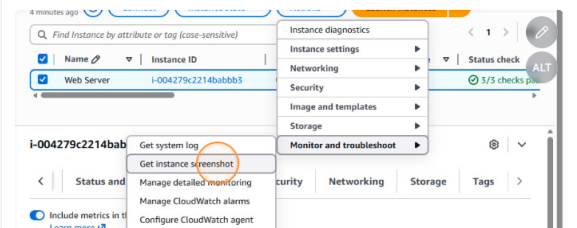
38 Click here.



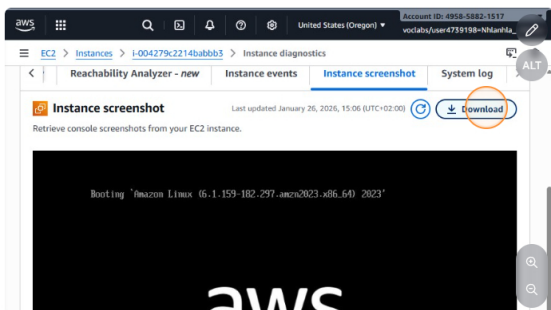
39 Click "Monitor and troubleshoot"



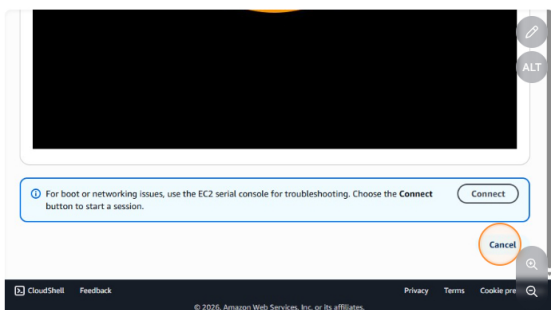
40 Click "Get instance screenshot"



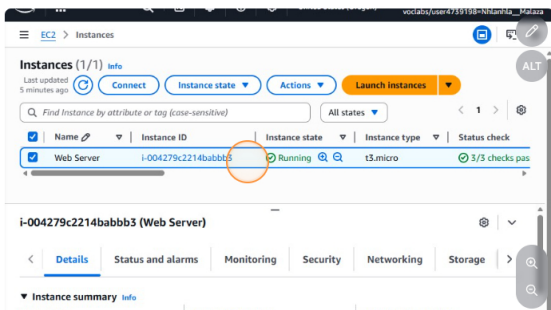
41 Click "Download"



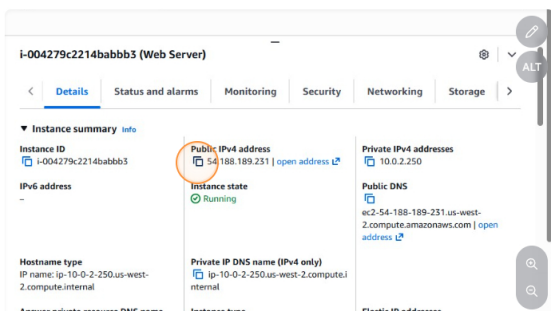
42 Click "Cancel"



43 Click "i-004279c2214babbb3"



44 Click here.



45 In a new tab, navigate to <http://54.188.189.231/>

46 Click "Connection timed out"

ALT



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47 Switch to tab Instances | EC2 | us-west-2"

+

48 Click here.

+

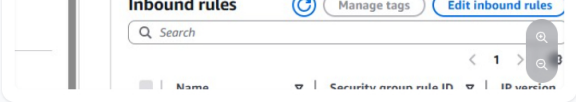
49 Click "Security Groups"

+

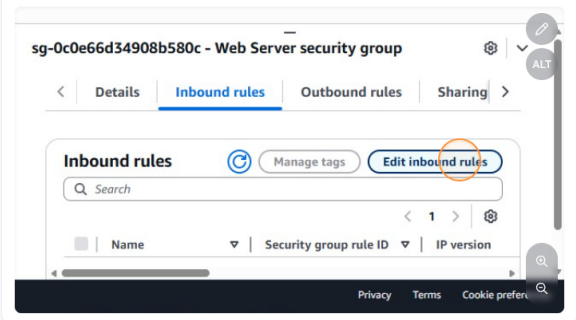
50 Click this checkbox.

+

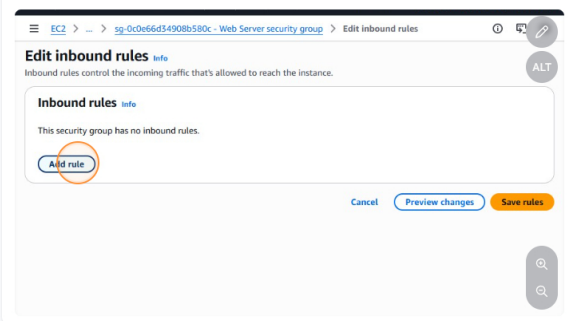
51 Click "Inbound rules"



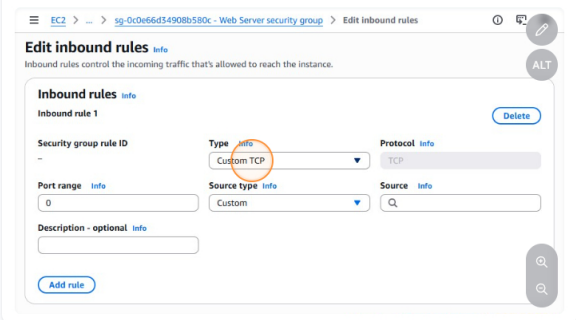
52 Click "Edit inbound rules"



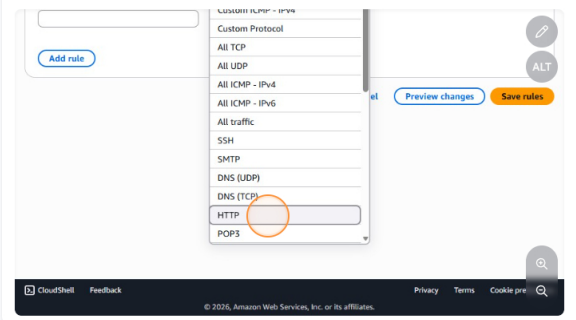
53 Click "Add rule"



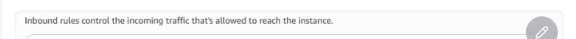
54 Click "Custom TCP"



55 Click "HTTP"



56 Click "Custom"



Security group rule ID

Type [Info](#)

Protocol [Info](#)

Port range [Info](#)

Source type [Info](#)

Source [Info](#)

Description - optional [Info](#)

[Add rule](#)

[Cancel](#) [Preview changes](#) [Save rules](#)

57 Click "Anywhere-IPv4"

Inbound rules [Info](#)

Inbound rule 1

Security group rule ID

Type [Info](#)

Protocol [Info](#)

Port range [Info](#)

Source type [Info](#)

Source [Info](#)

Description - optional [Info](#)

[Add rule](#)

[Delete](#)

[Cancel](#) [Preview changes](#) [Save rules](#)

58 Click "Save rules"

Description - optional [Info](#)

[Add rule](#)

[Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.](#)

[Cancel](#) [Preview changes](#) [Save rules](#)

[CloudShell](#) [Feedback](#) [Privacy](#) [Terms](#) [Cookie pre](#)

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59 Click "Details"

Account ID: 4958-5882-1517

United States (Oregon)

[Inbound security group rules successfully modified on security group \(sg-0c0e66d34908b580c | Web Server security group\)](#)

[Details](#)

Security Groups (3) [Info](#)

[Actions](#) [Export security groups to CSV](#) [Create security group](#)

[Name](#) [Security group ID](#) [Security](#)

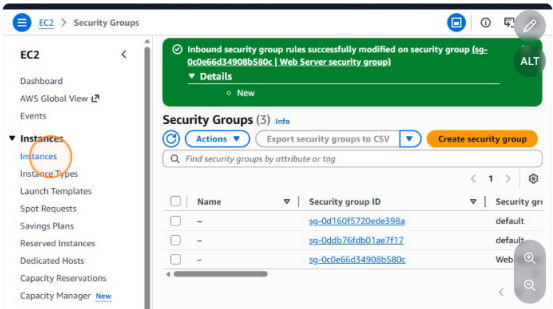
60 Type "F5"

61 Click "Hello From Your Web Server!"

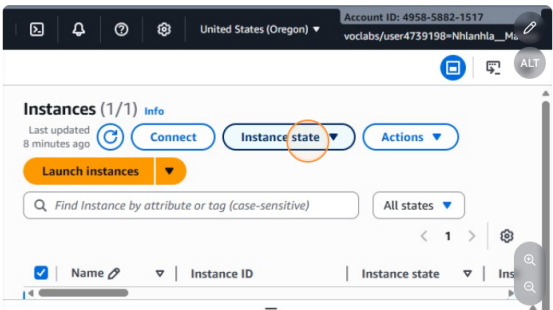
Hello From Your Web Server!

62 Switch to tab SecurityGroups | EC2 | us-west-2"

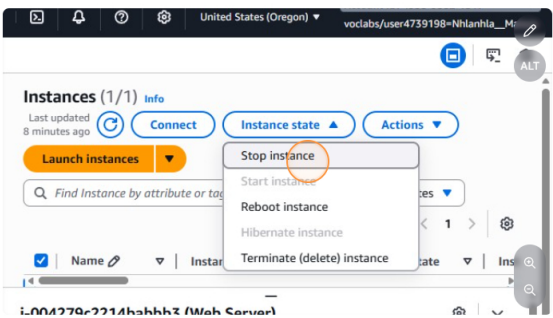
63 Click "Instances"



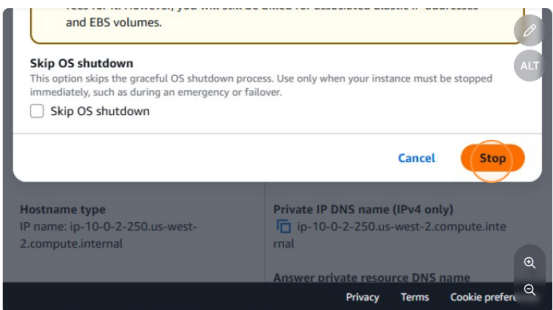
64 Click "Instance state"



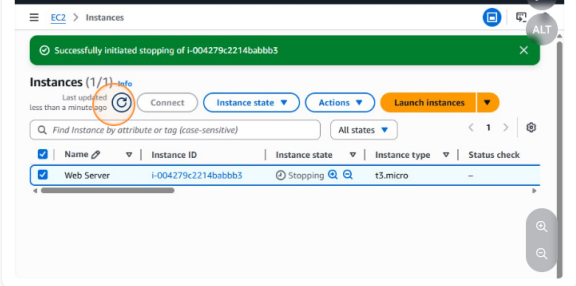
65 Click "Stop instance"



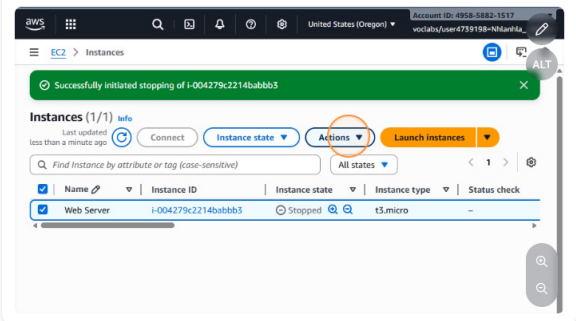
66 Click "Stop"



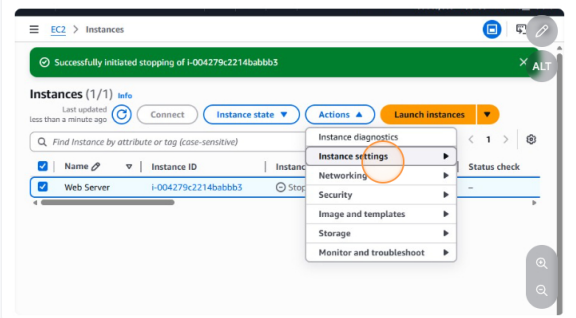
67 Click this button.



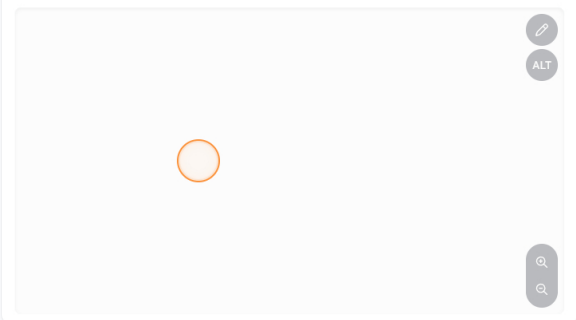
68 Click "Actions"



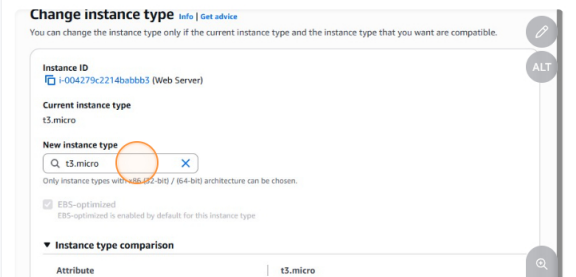
69 Click "Instance settings"



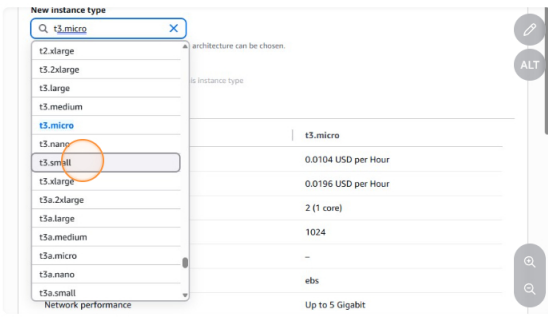
70 Click "Change instance type"



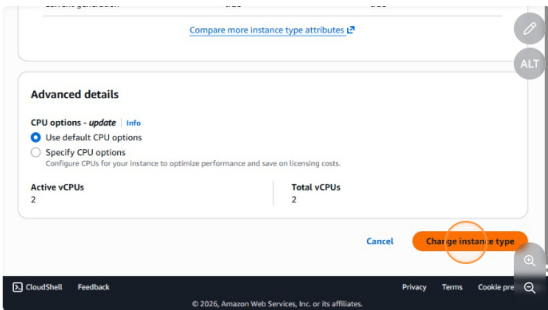
71 Click the "New instance type" field.



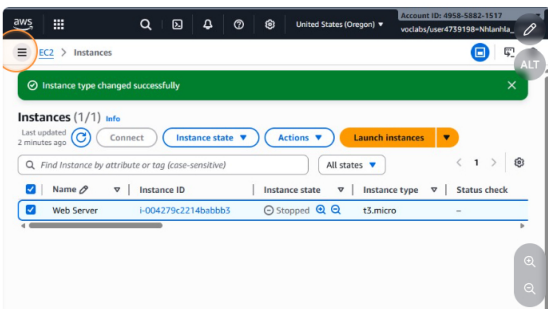
72 Click "t3.small"



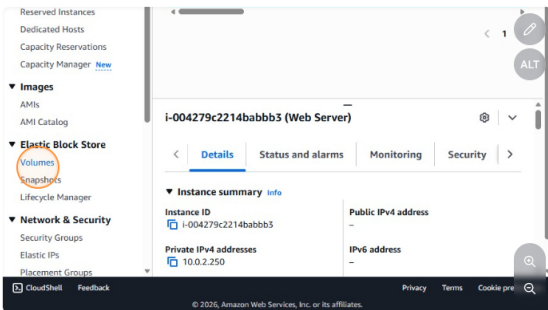
73 Click "Change instance type"



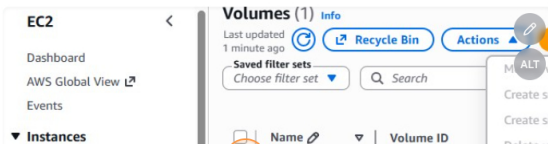
74 Click this button.



75 Click "Volumes"



76 Click this checkbox.



Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts

Force det

Ma
e i
Co
li
Ma
e i

+

77 Click "Actions"

The screenshot shows the AWS Management Console interface for the 'Volumes' section. At the top, there's a navigation bar with icons for home, notifications, help, and settings, followed by the region 'United States (Oregon)' and account information. Below this, the 'Volumes (1/1)' section is displayed. It includes a 'Last updated' timestamp, a 'Recycle Bin' button, and an 'Actions' dropdown menu which is highlighted with a red circle. To the right of the 'Actions' menu is a 'Create volume' button. Below these elements is a search bar and a table of volumes. The table has columns for 'Name', 'Volume ID', 'Type', and 'Size'. One volume is listed with ID 'vol-011a4baa332c3e962' and size '8 GiB'.

+

78 Click "Modify volume"

This screenshot shows the same AWS Management Console interface as the previous one, but with the 'Actions' dropdown menu open. The 'Modify volume' option is highlighted with a red circle. Other options visible in the dropdown include 'Create snapshot', 'Create snapshot lifecycle policy', 'Delete volume', 'Attach volume', 'Detach volume', and 'Force detach volume'. The volume table below still shows the same volume.

+

79 Click the "Size (GiB)" field.

The screenshot displays the 'Volume details' page for the volume 'vol-011a4baa332c3e962'. It shows the 'Volume type' as 'General Purpose SSD (gp3)'. The 'Size (GiB)' field is highlighted with a red circle and contains the value '8'. Below this field, it indicates 'Min: 1 GiB, Max: 65536 GiB'. Other fields shown include 'IOPS' (3000) and 'Throughput (MiB/s)' (125).

+

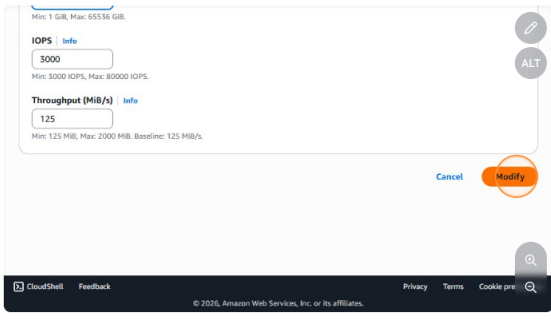
80 Type " Backspace 10"

+

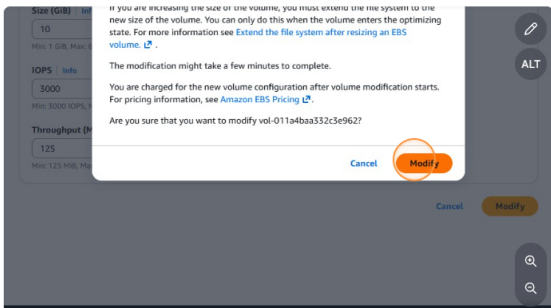
81 Click the "Size (GiB)" field.

This screenshot is similar to the previous one, showing the 'Volume details' page. The 'Size (GiB)' field is highlighted with a red circle and now contains the value '10'. The rest of the page content remains the same.

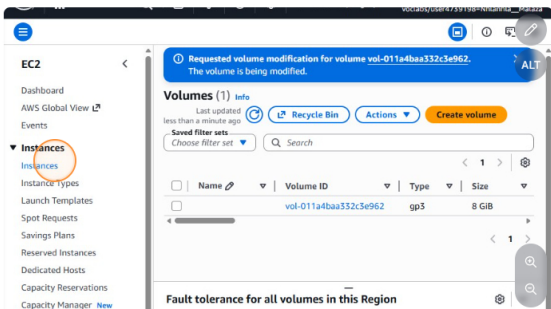
82 Click "Modify"



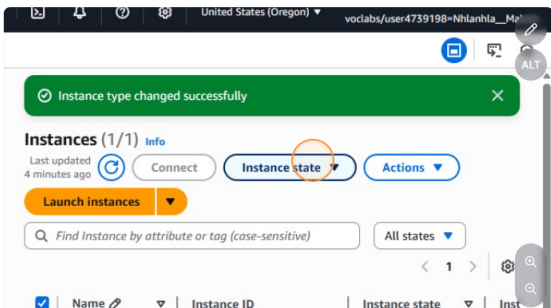
83 Click "Modify"



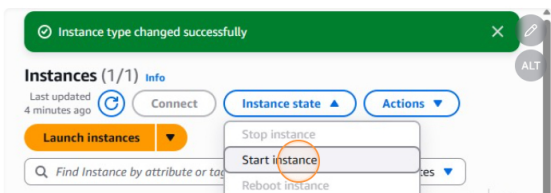
84 Click "Instances"



85 Click "Instance state"



86 Click "Start instance"



- 87 Click "Instances (1/1) InfoLast updatedless than a minute agoConnectInstance stateLaunch instancesAll states1"

The screenshot shows the AWS Management Console 'Instances' page. At the top, there's a header with 'Instances (1/1)' and 'Info'. Below this, there are buttons for 'Connect', 'Instance state', and 'Actions'. A search bar is present with the placeholder text 'Find Instance by attribute or tag (case-sensitive)'. Below the search bar, there's a table with columns: 'Instance state', 'Instance type', 'Status check', and 'Alarm status'. The first row shows a 'Running' instance of type 't3.small' with '2/2 checks passed'. The 'Instance state' dropdown menu is highlighted with an orange circle.

- 88 Click "Instance state"

The screenshot shows the AWS Management Console 'Instances' page. At the top, there's a green notification bar that says 'Successfully initiated starting of i-004279c2214babb3'. Below this, there's a 'Notifications' bar. The 'Instance state' dropdown menu is highlighted with an orange circle. The table below shows the instance is 'Running'.

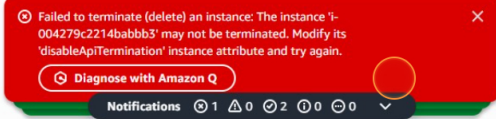
- 89 Click "Terminate (delete) instance"

The screenshot shows the AWS Management Console 'Instances' page. The 'Instance state' dropdown menu is open, and the 'Terminate (delete) instance' option is highlighted with an orange circle. The table below shows the instance is 'Running'.

- 90 Click "Terminate (delete)"

The screenshot shows the AWS Management Console 'Terminate (delete)' confirmation dialog. The 'Terminate (delete)' button is highlighted with an orange circle. The dialog contains the following text: 'To confirm that you want to delete the instances, choose the TERMINATE button below. Instances with termination protection enabled will not be terminated. Terminating the instance cannot be undone. Skip OS shutdown This option skips the graceful OS shutdown process. Use only when your instance must be stopped immediately, such as during an emergency or failover. Skip OS shutdown []'. At the bottom, there are buttons for 'Cancel' and 'Terminate (delete)'. Below the dialog, there's a table with instance details: Instance ID (i-004279c2214babb3), Public IPv4 address (16.145.8.37), Private IPv4 addresses (10.0.2.250), and IPv6 address (-).

- 91 Click "Failed to terminate (delete) an instance: The instance 'i-004279c2214babb3' may not be terminated. Modify its 'disableApiTermination' instance ..."



Instances (1/1) Info

Last updated
1 minute ago



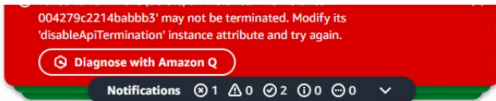
Connect

Instance state ▼

Actions ▼

Launch instances ▼

92 Click "Actions"



Instances (1/1) Info

Last updated
1 minute ago



Connect

Instance state ▼

Actions ▼

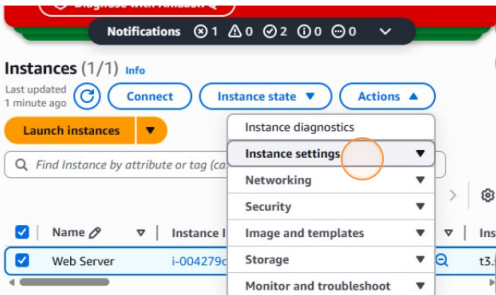
Launch instances ▼

Find Instance by attribute or tag (case-sensitive)

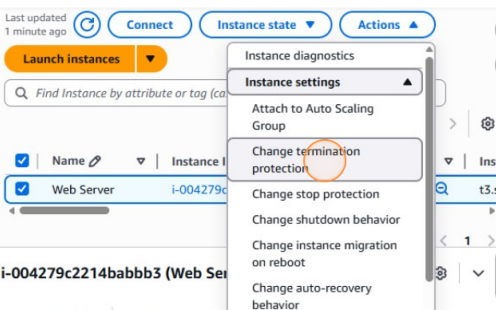
All states ▼

✓ Name Instance ID Instance state

93 Click "Instance settings"



94 Click "Change termination protection"



95 Click "Enable"

Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager [New](#)

Images

AMIs
AMI Catalog

Elastic Block Store

Volumes
Snapshots

Change termination (deletion) protection

To prevent your instance from being accidentally deleted, you can enable termination protection for the instance. [Learn more](#)

Instance ID

i-004279c2214babbb3 (Web Server)

Termination protection

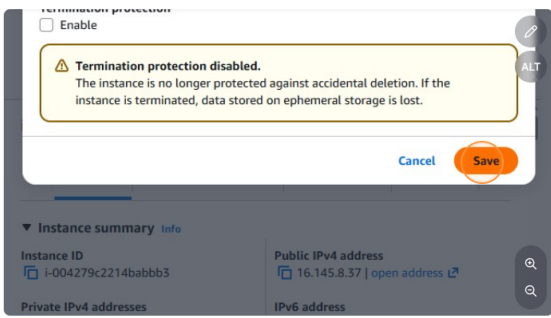
✓ Enable

Details

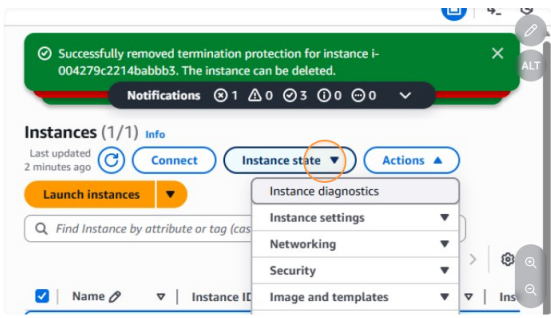
Status and alarms

More

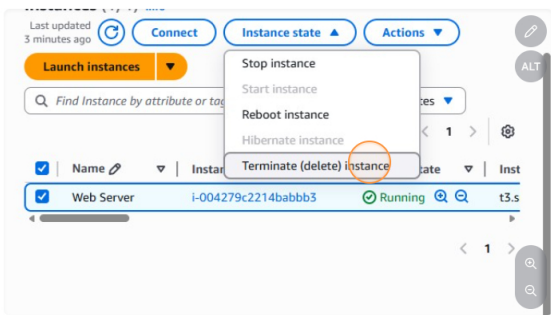
96 Click "Save"



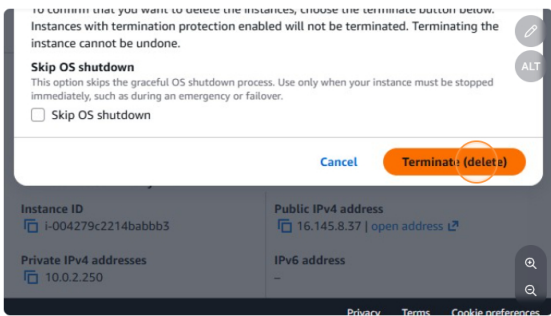
97 Click "Instance state"



98 Click "Terminate (delete) instance"

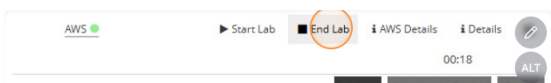


99 Click "Terminate (delete)"



100 Switch to tab Workbench-Vocareum"

101 Click "End Lab"



54. In the **Actions** menu, select **Instance State** ▶
Terminate instance.

55. Select **Terminate**

👏 **Congratulations!** You have successfully tested termination protection and terminated your instance.



102 Click "Yes"

